

Embracing Progressive Delivery in Kubernetes with GitOps

GitOps leverages Git's strengths to achieve reliable, automated, and consistent infrastructure and application management, making it well-suited for modern DevOps practices and containerised environments like Kubernetes.



Kubernetes is an open source container orchestration platform that automates the deployment, scaling, and management of containerised applications. It provides a powerful and flexible solution for managing containerised workloads in a highly efficient and scalable manner.

Kubernetes plays a central role in modern cloud-native application development and has become the *de facto* standard for container orchestration due to its powerful features, scalability, and wide adoption within the industry.

Key features of Kubernetes

Here are the key features that distinguish the versatility of this platform.

Container orchestration:

Kubernetes manages the deployment and operation of containers, allowing

developers to focus on application logic rather than infrastructure details.

Automatic scaling: Kubernetes can automatically scale the number of container replicas based on defined resource utilisation metrics, ensuring applications are highly available and can handle varying loads.

Service discovery and load balancing: Kubernetes provides built-in service discovery and load balancing mechanisms, making it easy for containers to communicate with each other and distribute incoming traffic.

Self-healing: Kubernetes continuously monitors the health of containers and can automatically restart or replace unhealthy containers to maintain application availability.

Rollouts and rollbacks: Kubernetes supports rolling updates, enabling seamless application updates without downtime. It also allows easy rollbacks

to a previous version in case of issues.

Storage orchestration: Kubernetes manages persistent storage for containers, enabling applications to store and access data that persists across container restarts.

Configurations and secrets management: Kubernetes allows you to define configuration data and secrets separately from the application code, promoting security and easy management.

Horizontal and vertical scaling: Kubernetes supports both horizontal scaling (increasing the number of replicas) and vertical scaling (increasing container resources like CPU and memory) to meet varying workload demands.

Batch processing and Cron jobs: Kubernetes provides support for batch processing workloads and scheduling periodic jobs using Cron expressions.